Fundamental Stock Selection on the Point-in-Time S&P 500: Bias Audit and Backtest

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Abstract

The course FinRL-Trading pipeline was run end to end, and the equal-weight baseline was reproduced: a final NAV of 1.31, against 2.55 for SPY. Further investigation shows that most of the gap to SPY came from the backtest, not from the stock selection itself. The main issue is that the backtest does not account for overnight returns. Three other problems also needed fixing: a survivorship leak in the price table, a warm-up period with almost no live signals, and two look-ahead leaks in the course code.

With all of them fixed, the strategy (the average of the three models' predictions, top 25%, rank-weighted, rebalanced monthly) ends at **NAV 2.563 vs 2.548 for SPY** over 2020-05-01 to 2025-09-30, net of 0.1% costs.

1 Data

This assignment uses publicly available data: daily prices for 501 tickers (2015–2025), quarterly financial reports turned into 52 ratios (price-to-earnings, margins, debt and so on) for 715 tickers, and a list of which companies were in the S&P 500 on each day since 1996. `prepare_data.py` keeps the 21,299 rows that have a return label. Only companies that were actually in the index on each date are kept: this point-in-time filter removes **2,292 rows and 18 tickers**, which leaves **19,007 rows, 674 companies and 42 quarters** (2015-06 to 2025-09).

Companies that later left the index stay in the sample for exactly the quarters they were members, for example Whole Foods, St. Jude Medical, Linear Technology, Activision, Xilinx, SVB Financial and First Republic. The models trade 22 out-of-sample quarters, and the backtest runs from 2020-05-01 to 2025-09-30.

2 Bias audit

2.1 Look-ahead bias

Look-ahead bias means the strategy uses information before it was publicly available. Signals are timed by each report's filing date rather than the quarter end, and each model is fitted only on earlier quarters, so model fitting is out of sample. However, two leaks were still hiding in the original pipeline:

- The first leak is in the valuation ratios. P/E and similar ratios use the stock price on the first trading day after the quarter ends. But some companies, like TJX, Oracle and Intuit, have fiscal years that do not follow the calendar and file their reports before that trading day. TJX filed on November 29, 2023, but its P/E uses the price from January 2, 2024, about a month later. In this pipeline, about 12% of predictions were affected. To solve this, the strategy waits until the price inside each ratio actually exists: a prediction is usable only at max(filing date, first business day after the quarter end).
- The second leak is in model selection. Each quarter, the pipeline keeps whichever of its three models was most accurate recently. But judging "most accurate" used returns that were not known until about two months after the quarter ended (the label `y_return` runs between "trade dates" two months after consecutive quarter ends), while the signal starts trading about one month after. To solve this issue, the strategy averages all three models instead of choosing one of them.

The strategy also trades one day after each signal becomes usable, so a report filed after the market closes is never traded before the market can react.

The four fully leak-free versions end between 2.50 and 2.56. The version that beats SPY was chosen after all four were seen, so that choice was made on results, and its 0.6% margin is far inside the spread between them.

2.2 Survivorship bias

Survivorship bias means testing a strategy only on the companies that still exist today, quietly removing every company that went bankrupt or got bought. The course's list of companies handles this correctly: it includes Whole Foods, SVB and Activision for

Signal version (same portfolio rule)	NAV	Beats SPY (2.548)?
Course pipeline (look-ahead)	2.560	yes, but uses look-ahead
Fix 1 only	2.522	no
Fix 2 only	2.475	no
Fix 1 + Fix 2	2.502	no
Fix 1 + average of 3 models	2.519	no
Fix 1 + Fix 2 + 1-day lag	2.527	no
Fix 1 + average + 1-day lag (used)	**2.563**	yes

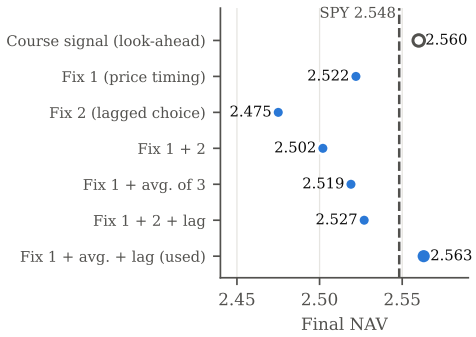


Figure 1: The same portfolio rule under each signal construction. Only one of the four leak-free versions ends above SPY.

the years they were in the index. But the price file only covers 501 of the 674 historical members, leaving out companies that are no longer in the index, and the notebook gives those 173 companies a 0% return (10.9% of the baseline's weight on average).

Simply skipping the stocks without prices is not a fix; it is the bias itself, and it would make the same strategy jump from $2.56 to $2.86. Instead, prices for 84 of the missing companies come from Yahoo Finance, including ones that only changed their ticker symbol. Each series is checked against the dataset's own prices, with two special cases:

- Paramount's old symbol now belongs to a different security, so it is dropped.
- SVB and First Republic are booked at a 100% loss on the day regulators seized them.

Eighteen held names (about 5% of the weight, mostly acquisitions such as ATVI and XLNX) still have no price anywhere and earn 0%. If they had earned their sectors' average return instead, the NAV would be 2.68 rather than 2.56, so 0% is probably conservative.

2.3 Warm-up

A stock only has a prediction once its company has filed a quarterly report. With the course's original timing, only one company (Textron) had a live prediction at the first rebalance on May 1, 2020, so the "top 25%" portfolio was 100% in that one stock until more companies filed. Those first three months returned +27.5% against +16.0% for the S&P 500, so a handful of lucky stocks drove most of the early lead. A rule was therefore added so the result does not depend on so few stocks: hold the index until at least half of the companies in the index have a live prediction (in practice May–July 2020). With the leak-free timing, no signal is usable before July 2020; without the rule, the strategy would sit in cash while the market rallied and end at NAV 2.20, so the rule also raises the final result. Any threshold from 10% to 90% gives a NAV between 2.56 and 2.61.

3 Stock selection

For each sector and each rolling window, three models (Random Forest, XGBoost and LightGBM) are tuned with grid search. In the course's original pipeline, XGBoost won 93 of the 238 sector-windows, Random Forest won 85 and LightGBM won 60, so no single model is consistently the most accurate. The strategy therefore averages the three instead of picking a single winner.

The course code does not save feature importances, so a proxy was trained: one pooled XGBoost model on the periods before 2020. The importances are almost flat (inventory turnover 2.5%, P/B 2.4%; a uniform split is 1.9%), which means that no single ratio dominates the predictions.

The signal is weak: the mean quarterly rank IC between predicted and actual returns is about 0.004 for the course's selected model and 0.020 ($t = 0.62$) for the three-model average used here. Neither is statistically different from zero. The strategy keeps the notebook's top-25% selection.

4 Strategy and backtest

On the first trading day of each month, at the close, the strategy ranks every live prediction across all sectors and keeps the top 25% of the companies, about 123 stocks.

Within that group, each stock is weighted by the rank of its prediction: the best prediction gets about twice the median weight, and the largest position averages 1.6%. Between rebalances, the weights drift with prices. The strategy assumes a trading cost of 0.1% of the amount traded; about a third of the portfolio is traded each month, so costs add up to 2.2% over the whole period.

The original notebook's backtest computes each stock's daily return as the closing price over the opening price, which means buying at the open and selling at the close of the same day. In reality, a typical portfolio holds stocks overnight and also earns the move from one day's close to the next day's open. The notebook's SPY benchmark, on the other hand, is measured from close to close, so the original comparison was not like for like.

How much does this matter? An equal-weight portfolio of every stock in the dataset over the same period shows the size of the effect:

- Counting only open-to-close moves, $1 grew to $1.26.
- Counting from close to close, it grew to $2.78, so in this sample more than half of the return came overnight.

The baseline therefore lost to the S&P 500 mainly because it missed overnight returns, not because of bad stock choices. The strategy and the benchmarks (SPY and QQQ total return, cached so the results are reproducible) all use close-to-close returns with dividends. Rebalancing monthly instead of re-weighting daily also roughly halves turnover (0.029 to 0.016 per day) and cuts costs from 3.9% to 2.2% in total.

5 Performance

2020-05 to 2025-09, net of 0.1% costs	NAV	Cum. return	CAGR	Vol	Sharpe	Max DD	Turnover/day
Baseline	1.31	+30.7%	5.1%	15.2%	0.40	−19.8%	0.032
The reported strategy	**2.563**	**+156.3%**	**19.0%**	20.7%	0.95	−31.4%	0.016
Top 25%, equal weight	2.561	+156.1%	19.0%	19.6%	0.99	−28.4%	0.016
Sector-neutral scoring	2.34	+134.0%	17.0%	18.2%	0.96	−26.5%	0.028
No-model control (all members, EW)	2.29	+128.9%	16.5%	16.5%	1.01	−20.1%	0.006
Survivors only (biased)	*2.86*	*+186.3%*	*21.4%*	*20.4%*	*1.06*	*−29.5%*	*0.016*
SPY	2.548	+154.8%	18.9%	17.5%	1.08	−24.5%	–
QQQ	2.91	+191.5%	21.8%	23.0%	0.98	−35.1%	–

Table 1: Leak-free signal and warm-up rule for every row except the reproduced baseline. Sharpe uses rf = 0.

With all the changes in place, the final strategy buys the top 25% of the averaged predictions each month and gives slightly more weight to the best ones.

From May 2020 to September 2025, after trading costs, $1 grew to **$2.563**, while SPY grew to **$2.548**. The strategy beats the index by about 1.5 cents on the dollar, a negligible advantage, and it takes more risk to get there: higher volatility, a deeper drawdown and a lower Sharpe ratio.

After the warm-up, the strategy is ahead of SPY on 85% of trading days; its widest deficit was 5%, in the April 2025 sell-off. Equal weighting does just as well with a better Sharpe, so the two rules cannot really be told apart.

On average, 64% of the portfolio's money was invested in technology stocks, against 15% when the money is spread evenly across every stock in the index. Each sector has its own model, and the technology model, trained on years when tech performed well, happens to predict larger returns than the other sectors' models. Ranking all sectors' predictions on one scale therefore puts mostly tech stocks at the top.

When each stock is ranked only against others in its own sector, the tech weight drops to 15% and the strategy ends at $2.34, below the index. Most of the strategy's advantage therefore came from the accidental bet on tech companies. Compared with the no-model control, the model adds 2.5 points of CAGR, but only about 0.5 once scoring is sector-neutral.

6 Robustness

Four checks show how fragile the lead is:

- **Split the period** – It won in 2020–22 (15.4% a year vs 13.7%) and lost in 2023–25 (22.7% vs 24.2%), when a few very large tech companies drove most of the index's gains.
- **Test the significance** – The excess return is about 0.7% a year with a t-statistic of 0.23, suggesting the gain is not statistically significant. Beta is 1.12, so the extra return is explained by extra market exposure.

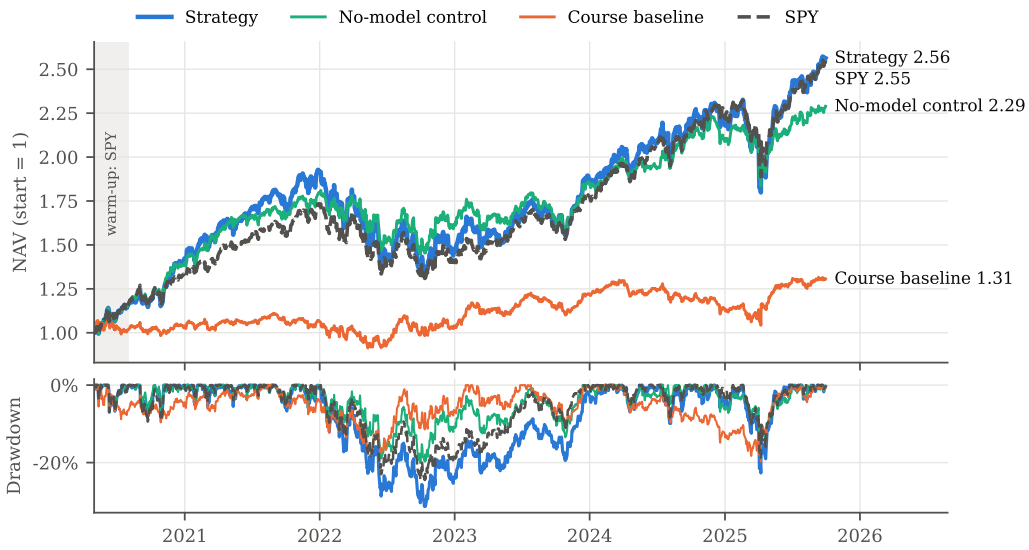


Figure 2: NAV (top) and drawdown (bottom), net of costs. The shaded band is the warm-up, when every strategy holds SPY.

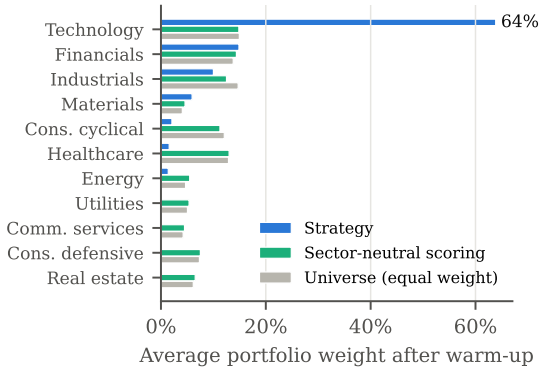


Figure 3: Average sector weights after the warm-up. Ranking predictions from 11 separately trained sector models together produces a 64% technology weight.

Check	Result
Sub-periods, CAGR (Sharpe)	2020–22: 15.4% (0.75) vs SPY 13.7% (0.76); 2023–25: 22.7% (1.18) vs SPY 24.2% (1.47)
Excess return vs SPY	+0.7%/yr, tracking error 7.1%, t = 0.23 (Newey–West)
Market exposure	beta 1.12, alpha −1.5%/yr (t = −0.54)
Multiple testing	Deflated Sharpe probability ≈ 3% after ∼50 variants
Transaction costs	NAV 2.62 / 2.56 / 2.51 / 2.35 at 0 / 0.1 / 0.2 / 0.5%; break-even ≈ 0.13%
Validation-preferred variants	volatility weighting 2.51, retrained model 2.48, momentum blend 2.22: all below SPY
Literature factors	7 anomalies, none significant; pre-2020 quality composite 2.26

Table 2: Robustness of the strategy. Validation = 2020-05 to 2022-12, test = 2023-01 to 2025-09.

- **Higher risk** – The portfolio is more volatile than the index (20.7% vs 17.5% a year) and had a deeper drawdown (−31% vs −24%). It also earns less return per unit of risk (a Sharpe ratio of 0.95 vs 1.08).
- **Higher costs** – At a 0.2% trading cost instead of 0.1%, the advantage is gone (break-even is about 0.13%).

7 Conclusion

The strategy meets the requirement (NAV 2.563 vs 2.548) on a sample audited for survivorship, look-ahead and warm-up bias. However, it is not worth investing in: the margin over the index is only about 1.5 points over the whole five-year period, and it is not statistically significant. The strategy is also sensitive to costs and depends on an accidental tilt toward technology stocks. The most useful outcomes of this assignment were the backtest fixes: the overnight-return fix alone explains most of the baseline's gap to SPY, and ignoring the missing prices would have inflated the NAV from 2.56 to 2.86.